Machine Learning Assignment

1. Problem Statement

Faclon Labs aims to build an advanced Power Demand Forecasting system to accurately predict future energy consumption based on real-time operational and sensor data. This assignment assesses your ability to design a robust, interpretable, and production-ready predictive solution that can drive energy optimization and operational efficiency.

Your objective is to create a machine learning model that forecasts Power Demand (Energy Consumption) using the provided dataset.

2. Data Overview

Dataset Summary:

- Total Records: 40,952
- Total Features: 23
- Target Variable: target (Energy Consumption)

Feature Categories: The dataset includes timestamp-based features, enabling trend analysis, seasonality detection, and operational correlation modeling.

Feature List : Provided in mail | Dataset : Provided in mail

3. Assignment Objectives

Design and develop a Power Demand Forecasting solution focused on:
1. Accuracy: Deliver precise energy consumption forecasts.
2. Robustness: Ensure reliability across unseen or changing operational conditions.
3. Explainability: Provide actionable insights into the most influential parameters.
4. Deployment Readiness: Outline an end-to-end production deployment strategy.

4. Expected Deliverables

A. Predictive Modeling:

- Build and train an optimized forecasting model for the target variable.
- Apply suitable feature engineering and validation methodology.
- Evaluate model performance using Mean Absolute Percentage Error (MAPE) as the primary metric.
- Document design choices and validation approach.

B. Explainability & Insights:

- Conduct feature importance and impact analysis to interpret model predictions.
- Use explainability techniques to show which variables influence energy demand most significantly.

C. Deployment & Integration Strategy:

- Provide a high-level deployment architecture suitable for IoT-driven production.
- Describe integration points with existing data systems, retraining cadence, and monitoring strategies.

5. Evaluation Criteria

- **Data Understanding & Preparation:** Exploration, preprocessing, and feature transformations.
- **Model Design & Optimization:** Justification of approach and validation methodology.
- **Performance Evaluation:** Accuracy, consistency, and generalizability (MAPE).
- **Explainability & Actionability:** Quality of insights from interpretability analysis.
- **Deployment Strategy:** Practicality and scalability of the proposed solution.
- **Code Quality:** Modularity, documentation, and coding standards.
- **Presentation:** Clarity, visuals, and communication of results.

6. Submission Requirements

1. Code Implementation:

- Format: Jupyter Notebook (.ipynb) or Python Script (.py).
- Well-documented code with preprocessing, modeling, and evaluation steps.

2. Presentation Deck:

- Format: PowerPoint (.pptx) or PDF.
- Include methodology, key findings, explainability visuals, and deployment overview.

7. Key Notes

- Demonstrate both strategic and technical thinking.
- Clearly document assumptions, rationale, and potential improvements.
- Focus on reliability, scalability, and maintainability for production use.

8. Submission Timeline

- Code & Presentation Submission: Within 72 hours of receiving dataset

Confidentiality Notice:

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